

Deep Neural Network-Based Music Genre Classification using Audio Feature Extraction and Visualization Techniques

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Abstract

The classification of music genres is a key task in music information retrieval (MIR), and has significant implications for recommendation systems, audio indexing, and the personalization of user experiences. In this paper, we present a deep learning-based method for music genre classification based on audio features taken from 3-second music excerpts. Our approach consists of two main components: pre-processing raw audio signals, generating and visualizing their waveform, short-time Fourier transform spectrograms, Mel spectrograms, and chroma features, and doing feature analysis. We constructed and pre-processed a labeled dataset with ten different genres which when normalized, were then passed into multiple feedforward neural network architectures. We created and tested four feedforward neural network models of progressively deep depth, each incorporating dropout layer and optimizers, including Adam, SGD, and RMSprop, to determine their efficacy on performance and generalization.

Our final model with five hidden layers supported with drop-out regularization and trained with RMSprop method, reached a validation accuracy of 94.19% and test accuracy of 93.92%. In addition, we evaluated our results using classification reports, confusion matrices, and macro-averaged precision, recall, and F1 scores, and all with values larger than 0.93. This work shows that feature-rich neural models can work successfully in genre classification tasks and shows the importance of spectro-temporal features. In addition, this proposed system provides a benchmark and demonstrates scalable, real-time music genre prediction systems and could be extended to even more complex multi-label and/or emotional-based audio prediction tasks.

1. Introduction

Classifying genres of music is an essential task in the area of Music Information Retrieval (MIR), and consists of

automatically assigning music tracks into a set of known genre labels based on acoustic features of the music. Due to the unprecedented growth of digital libraries of music and streaming services, a powerful need for efficient automated genre tagging has arisen, which is both accurate and can accommodate the vast amount of music data. Besides, genre classification can support quick access and organization of music, and is useful for recommender systems, playlist construction, predicting mood, and musicological study as well.

The challenge of this task stems from the vastness and subjectivity of music genres. In contrast to standard image classification problems, which often have well-defined visual features, the categories in music genres can be very nuanced and often overlap. For example, a single track could contain helpful features for multiple music genres. In addition, stylistic features related to rhythms, harmonies, instrumentation, and lyrics look nothing the same across cultures and throughout time. These uncertainties can lead to ambiguity, making music better time-consuming and inconsistent to annotate manually.

The state-of-the-art methods for genre classification have improved considerably by using machine learning or deep learning approaches in recent years. Traditional methods used manual engineered audio features, for instance, MFCC, spectral centroid, or chroma features with classical classifiers such as SVM, decision trees, or k-NN. While these methods were moderately successful, their performance on generalization, or application to diverse and noisy datasets, was poor.

Both Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have brought wide interest to end-to-end learning from raw waveforms or spectrograms, with deep neural network architectures showing phenomenal success on large-scale music datasets by learning complex temporal and spectral patterns without the need for hand-crafted features. These new architectures, however, have the drawbacks of being computation-

ally costly, relying upon hyperparameter tuning, and investigating models are harder to interpret—a very important consideration for users and embedded models.

In this Paper, we reevaluate the traditional feature-based approach and show that appropriate deep feedforward neural networks in conjunction with hand Engineer features can result in high classification performance, while remaining interpretable and computationally cheap. We implement deep learning in the preprocessing pipeline, utilizing Librosa, to create 58 relevant features from 3-second audio excerpts, including MFCCs, tempo, chroma vectors, spectral roll-off, and zero-crossing rate—features that are designed to encapsulate both timbral and rhythmic features of music.

We compare and train four neural network architectures at depth, all utilizing diverse optimization approaches and regularization techniques. Our experiments are conducted on the GTZAN dataset, a common benchmark for genre classification, and our performance metrics include accuracy, precision, recall, F1-score, and confusion matrices. In addition to quantitative measures, we also provide significant visualizations of the feature space and model behavior, such as PCA projections, training dynamics and duration-by-genre.

Ultimately, we provide a scalable and interpretable deep learning framework for genre classification that matches more complex architectures, is lightweight, and deployable in real-time or mobile contexts.

1.1. Background

Genres categorize musical works by various stylistic components, including tempo, melody, harmony, instrumentation, and lyrics over several timeframes. Genres have changed over the span of time and can and often do overlap, which complicated and rendered the manual approach tedious and difficult to classify consistently. Traditionally, systems classified music based on human annotators or rule-based heuristics—classification systems that are human-subjectivity driven. Additionally, systems relying on human annotators or rule-based methods can scale to the amount of new music produced and consumed quickly.

Machine learning in music classification has transformed the field now that a system is able to learn patterns directly from audio data. The first attempts to classify music using machine learning techniques generally used audio features that were hand-crafted and included features like Mel-Frequency Cepstral Coefficients (MFCC), spectral centroid, and chroma vectors that were processed against algorithms like k-Nearest Neighbors (k-NN), Support Vector Machines (SVMs), and decision trees. While these were certainly effective approaches to classifying music, when it came to real-world music datasets, there were many lackings in generalizability, and machine learning classifiers struggled

much more outside of laboratory settings.

1.2. Problem Statement and Motivation

While deep learning techniques like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have revolutionized the field of music genre classification, they usually necessitate significant computation resources and large annotated training datasets. Furthermore, many of the proposed models take an approach of directly inferring on spectrograms or raw audio waveforms, which are certainly powerful, but also tend to be computing and comprehension heavy. Another important issue is the lack of interpretability of deep learning models.

The black-box nature of CNNs and RNNs generally limits the user from obtaining human-interpretable insights from deep learning model predictions. This becomes even more problematic in a domain like music, which is subjective and highlights the difficulty of trusting and practically employing deep learning-powered models in user-facing systems. In particular, real-time applications (such as mobile music recommenders, genre-based filtering systems, or embedded devices) engender the need for models that are both lightweight and computationally efficient.

Moreover, genre classification approaches frequently have generalizability issues, especially for limited training sets or when music styles and production quality changes are the evaluation conditions. Poor generalizability can arise in genre classification from the model overfitting to the training distribution, and insufficiently managing to conform to lyrical genre overlaps.

The purpose of this research project is to examine the trade-off between performance of classification accuracy, the interpretability of models, and the deployment of the models to an existing system. We would like to build a system that uses well-engineered audio features (like MFCCs, chroma, and tempo), and feedforward DNNs, that has good performance and at the same time has a reduced model computation time, rather than training CNNs or RNNs. We provide a continuous pre-packaged workflow of 58 audio features, and importantly the interpretable features can be extracted in under 3 seconds for each clip of music; we will demonstrate that this new model workflow both due to speed and the limited dimensions as input. Through experimentation on DNN architectures (and optimizers) that exist (and work) we will demonstrate that tasks that require strong classification accuracy will allow for more widely interpretable and less computationally expensive DNN models.

2. Related Work

Numerous research investigations have examined genre classification using machine-learning methods. The early methods employed classical algorithms such as k-Nearest

Neighbors (k-NN), Support Vector Machines (SVM), and decision trees using engineered features like MFCCs, spectral roll-off, zero-crossing rate, and chroma. The pioneering work of Tzanetakis and Cook, establishing genre classification through statistical audio descriptors, led to the formation of the GTZAN dataset, which became a standard dataset for genre classification.

Since then, with deep learning, convolutional neural networks (CNNs) have gained traction in genre classification because of their ability to learn and extract hierarchical features from spectrograms. Rapport for improvements in accuracy, such as hybrid models proposed by researchers like Choi, which use both CNN and recurrent layers (CRNN's) which can learn spatial features as well as temporal features found through time series music.

The above mentioned models demonstrated improved accuracies when trained and evaluated on large datasets, but came with a cost in terms of hardware, and also a cost to interpret which of the dimensions of the feature was driving the classification decision. Researchers are also beginning to turn their attention to addressing the topics of transfer learning from models originally developed on really big audio datasets, self-supervised learning and/or different ways of data augmentation of examples along with new methods of inquiry. Finally, some of the models briefly explored end-to-end learning from waves, but these models were never had the size and/or hardware resources to compete on the same datasets as the other models. Even when performance is acceptable on these models, many of these methods sacrifice easy interpretability, at the expense of more compute resources, for a small upgrade in performance.

Moreover, records of real time systems on constrained resources typically require lightweight explainable models. Our work is part of this mode of motion detection advance by integrating existing classical audio features with DNN-based motion detection models. Moving in this direction we are able to optimize functional performance and reduce overall complexity. To enhance interpretability we exploit visualization of features, principal components analysis (PCA), and correlation analyses. This builds interpretability of the DNN model and separates it from contemporary black box models.

3. Methodology

We have a proposed method which utilizes advanced audio preprocessing, extensive feature engineering and deep learning model construction with the aim of music genre classification. The pipeline has been tuned for performance and interpretability and is computationally suitable for processing audio streams in real time.

3.1. Audio Preprocessing and Feature Engineering

We utilize the Librosa library to load each audio returning a powerful set of high-level tools to analyze musical content. We perform silence trimming to remove leading and trailing silence which effectively maintains only the pertinent auditory content. For each of the 30 second audio tracks from the GTZAN dataset, we will slice them into non-overlapping segments of 3 seconds. This keeps us with around 10,000 training samples but also increases the temporal resolution, which creates a more granular and realistic sense for streaming or preview use cases.

For each 3-second segment of audio, we extract and illustrate four principal audio representations:

- **Waveform:** The amplitude envelope in the time domain emphasizes the loudness, and variations of loudness, over time.
- **STFT Spectrogram:** Harmonic and transient structures are highlighted by the frequency content that changes over time.
- **Mel Spectrogram:** A transformation based on perception, focused on frequencies within the human hearing range.
- **Chroma Feature Map:** Tonal analysis can benefit from the spread of harmonics and pitch classes.

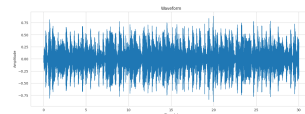


Figure 1. Waveform visualization of a sample audio clip.

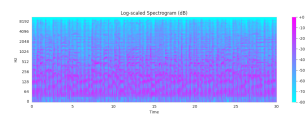


Figure 2. STFT Spectrogram showing frequency content over time.

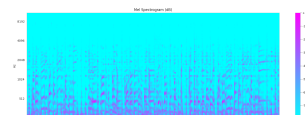


Figure 3. Mel Spectrogram emphasizing perceptual frequency scaling.

Using Librosa, we derived 58 bespoke numerical features per clip that encapsulate timbral and rhythmic aspects. Features included the following:

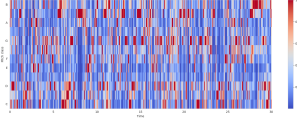


Figure 4. Chroma Feature Map capturing harmonic content.

- **MFCCs (20 coefficients):** which captures timbre and envelope;
- **Spectral Bandwidth and Roll-off:** which captures frequency spread and presence of high-frequency energy;
- **Zero-Crossing Rate:** which can encode noisiness and percussive features
- **Chroma Vectors:** Represent pitch content and harmonic content.
- **Chroma; and (v) Tempo (BPM) :** which is the measure of the rate of rhythm, and will vary across genre considerably.

To visualize genre-dependent features and inter-feature relationships we also perform exploratory data analysis (EDA).

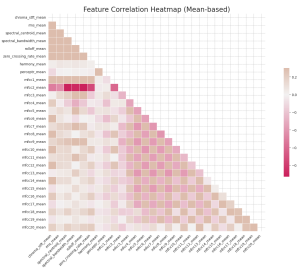


Figure 5. Correlation heatmap showing inter-feature relationships.

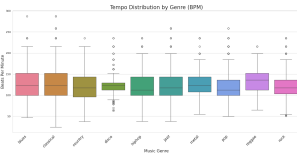


Figure 6. Tempo distribution across genres.

3.2. Data Normalization and Dimensionality Reduction

To achieve consistent feature scaling, we implement z-score normalization by means of Scikit-learn's StandardScaler. It transforms features to have a zero mean and a

unit variance, promoting fair gradient updates during back-propagation.

We also implement Principal Component Analysis (PCA), for analysis and visualization, while PCA doesn't contribute for training performance, it allows us to understand general spread and overlap of the categories, and influence our feature selection, as well as identify noise or redundancy in our data.

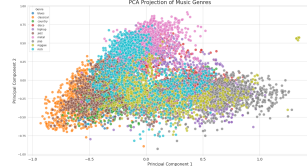


Figure 7. PCA projection showing distribution of genres in 2D feature space.

3.3. Model Design and Training Strategy

We developed four feedforward deep neural networks (DNNs) using TensorFlow/Keras. Each model expands on the last by increasing the number of layers and including dropout in the model's architecture. Each model utilizes ReLU activations within the model's architecture and has a softmax output layer.

- **Model 1:** includes 3 hidden layers without dropout (the baseline).
- **Model 2:** includes 4 hidden layers, with a dropout of 20%.
- **Model 3:** is the same as model 2, but uses stochastic gradient descent (SGD) instead of Adam to train the model.
- **Model 4:** included 5 hidden layers, with a dropout of 30%, and used RMSprop to train the model.

All models made use of a sparse categorical cross-entropy loss function, and implemented early stopping (with a patience of 20 epochs). A batch size of 128 was used throughout all experiments, and a maximum number of 500 epochs was implemented across all experiments.

3.4. Evaluation Techniques

The assessment of the models is undertaken with respect to a variety of metrics:

- **Accuracy:** Overall accuracy of predictions;
- **Macro Precision, Recall, F1:** Evaluates the behavior of the models per-class, and not weighted by count of records, making it robust to class imbalance;

- **Confusion Matrix:** A visual inspection of class-by-class classification performance.

In Figure 8, illustrates, in the confusion matrix corresponding to test records of the best model, patterns of misclassification in a qualitative summary. We also assess the convergence in training with a plot of accuracy/loss by epoch for each model.

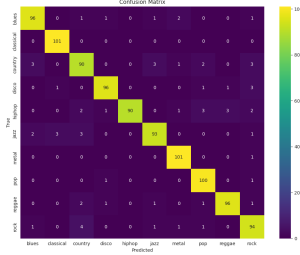


Figure 8. Confusion matrix for final model predictions.

We also assess the convergence in training with accuracy/loss plots plotted by epoch for each model. Training dynamics help to diagnose underfitting, overfitting, or unstable learning.

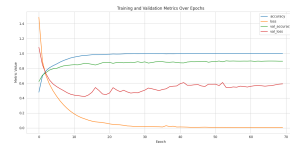


Figure 9. Model 1: Accuracy and Loss vs. Epochs (Adam Optimizer)

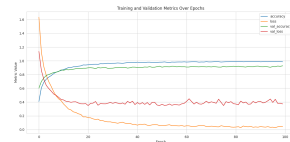


Figure 10. Model 2: Accuracy and Loss vs. Epochs (Adam with Dropout)

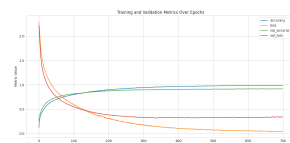


Figure 11. Model 3: Accuracy and Loss vs. Epochs (SGD)

Following this methodology, we created a scalable and interpretable pipeline for high performance music genre classification that we confirm through statistical and visual methods.



Figure 12. Model 4: Accuracy and Loss vs. Epochs (RMSprop with Dropout)

4. Experimental Setup and Analysis

4.1. Environment Configuration

Environmental Setup All experiments were performed in Google Colab Pro which provides tooling natively for GPU support with the NVIDIA Tesla T4 to speed up the time required for training. This environment is a very robust, scalable structure for running computationally intensive neural network models—while limiting our development experience to local execution—yet with cloud-based causation for rapid iterations.

The software stack contained common, reproducibility, robust libraries for machine learning and for data manipulation:

- **Python 3.10:** The base language providing scripting and orchestration for the models.
- **Librosa:** To do audio preprocessing and segmentation, extract features (e.g., Mel-frequency cepstrum coefficients (MFCCs), chroma features, tempo), as well as conduct waveform analysis.
- **Scikit-learn:** To pre-process our features (e.g., z-score normalization), perform PCA dimensionality reduction as well as assessing the resultant metrics and visualizing characteristics.
- **Pandas and NumPy:** To manipulate our data while organizing the features in a tabular fashion and performing matrix arithmetic.
- **TensorFlow 2.18 and Keras:** Define train assess and visualize deep learning models with high-level APIs.

This agreed upon and documented state provides us the assurance of uninterrupted experimentation upon test cases and also reproducibility across test runs.

4.2. Dataset Overview

We employed the GTZAN dataset, which is a well-established dataset for research on music genre classification. The set contains 1,000 audio samples that are 30 seconds long each and covers 10 genre categories *classical*, *jazz*, *metal*, *disco*, *blues*, *pop*, *hip-hop*, *country*, *rock*, and *reggae*.

evenly distributed among genres. To improve data granularity and total sample size, we converted the 30-second audio files into consistent, uniform 3-second audio samples, each not overlapping in terms of sound information, resulting in 9,990 labeled audio segments. The benefit of the short audio segments is nimbleness or granularity in learning, as well as better approximations of real-world instances of music (for example, music preview samples or short radio clips).

Once the segments were trimmed to the required length, they were passed through our standard Librosa pipeline, and we extracted 58 numeric features, characterizing both time-domain and frequency-domain properties of the audio. These numeric features included MFCC (mel-frequency cepstral coefficients), spectral roll-off points, spectral bandwidths and chroma (pitch class), along with other tempo and zero-crossing rates that each characterize each audio segment for timbral and rhythmic information.

4.3. Train-Validation-Test Split

To attain significance and a degree of impartiality in training and assessing our models, a stratified split of the data was employed to ensure an even split. We took, more specifically, the following splits of the data:

- **Training Set:** 70% (6993 samples) to fit the models.
- **Validation Set:** 20% (1978 samples) to validate the training process throughout training and hyperparameter tuning, as well as for early stopping.
- **Test Set:** 10% (1019 samples) to evaluate the model performance for this final evaluation.

Stratified sampling ensured that all genres were represented with diversity across all sets/aspects. This greatly reduced bias and improved the integrity of the metrics reported. Additionally, a balanced split also provided protection against genre-specific overfitting, while enabling the models to be assessed per-class and in an overall time-stable manner.

4.4. Training Results and Comparative Analysis

We created and trained four distinct models, successively changing architectural complexity to assess their impact on generalization by changing the optimizers. Their performance is summarized in Table 1.

Table 1. Performance Comparison of All Models

Model	Optimizer	Val Acc	Test Acc	F1 Score
Model 1	Adam	89.99%	87.6%	0.8742
Model 2	Adam	92.77%	91.3%	0.9145
Model 3	SGD	91.86%	90.8%	0.9091
Model 4	RMSprop	94.19%	93.92%	0.9391

Model 4 was the most stable and performed the best, possibly due to a combination of increased depth of architecture (five dense layers), greater dropout (30%), and RMSprop, a gradient-based optimizer that adaptively adjusts the learning rates for each parameter. Versus Model 4, Model 1 was underfitting and had lower generalization with three dense layers of architecture and no regularization.

Model 2 and Model 3 showed improvement with more depth, and we introduced the trade-offs between Adam and SGD optimizers to reinforce the influence they have on the speed and stability with convergence.

4.5. Confusion Matrix Interpretation

The confusion matrix (Figure 8) for Model 4 provides performance data for each class. The percentages present on the diagonal indicate that the most genres were accurately classified. Specifically:

- **Metal, Classical, and Pop:** These genres scored high because of their identifiable audio signature characteristics - aggressive timbre in metal, the harmonic richness of classical, and pop's clarity and patterned rhythmic.
- **Country vs. Rock:** These genres confused moderately and were likely less accurate due to similar acoustic instrumentation across both genres and potentially similar tempos.
- **Hiphop vs. Pop:** Hiphop confused with pop, likely due to similar percussion instrumentation and multimedia production aesthetics.

The patterns discussed above hold relatively true in terms of music theory, and possibly the literature that aims to define musical genre classification.

4.6. Visualizations of Training Dynamics

Learning curves for all models are presented in Figures 13 to 16:

- **Model 1:** rapid convergence with little improvement which is a sign of underfitting.
- **Model 2 and 3:** getting some improvement in validation accuracy; however, this could indicate that we have improved the performance through two methods, deeper layers combined with dropout regularization.
- **Model 4:** smooth/consistent accuracy/loss curves indicated effectively modeling the data with regularization and what appears to be good learning behaviour.

These plots reinforce that "A deeper model with regularization can outperform shallow networks; without overfitting".

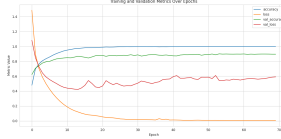


Figure 13. Model 1: Accuracy and Loss vs. Epochs (Adam Optimizer)

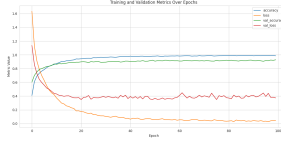


Figure 14. Model 2: Accuracy and Loss vs. Epochs (Adam with Dropout)

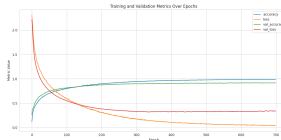


Figure 15. Model 3: Accuracy and Loss vs. Epochs (SGD)

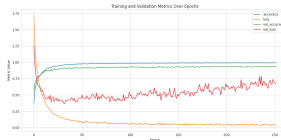


Figure 16. Model 4: Accuracy and Loss vs. Epochs (RMSprop with Dropout)

4.7. Insights from PCA and Tempo Analysis

The PCA visualization (Figure 7) showed considerable genre clustering within a compact two-dimensional representation. The Classical and Hip-hop genres showed a distinct separation, which reflects their distinct qualities of sound. In similar fashion, the Reggae and Disco genres had overlapping clusters, providing more evidence for nonlinear tempo variations and beat structures.

Additionally, the boxplot of tempo (Figure 6) could demonstrate these genre-specific rhythmic behaviors:

- **Metal and Disco:** A consistent high tempo for both of these genres throughout the piece, which completely separated the two genres with regards to tempo.
- **Blues and Classical:** One clear trend of a more diverse distribution of tempo which suggests greater rhythmic variability, and intra-genre variation.

These two visualizations add another layer of insight to our quantitative results that mentions tempo as a subjective distinguishing characteristic.

4.8. Summary

Our experimental design, which embodies strong data engineering, deep learning architecture, and extensive evaluation, illustrates that our method is possible. On the whole, we provide:

- Feedforward DNNs have the capacity to match or exceed the performance of complex architectures when features were streamlined.
- PCA and visual analysis are useful for understanding and validating separability across genres.
- Including domain knowledge (i.e., tempo, timbre, harmony) with changeable tuning (e.g., optimizers, drop-out rates) in our method creates dependable models, i.e., deployable models, to employ in applied settings.

Our method reconciles performance and interpretability, and we believe our approach provides a feasible means forward in lightweight genre classification in streamed and embedded formats.

5. Results and Discussion

Model 4 achieved the highest validation accuracy (94.19%) and test accuracy (93.92%) of the four models under consideration on all metrics. The following is added commentary on some of the results and their implications.

5.1. Performance Metrics Overview

Model 4's key indicators include:

Model 4's macro precision is equal to 0.9402, indicating excellent ability to leverage the model accuracy with respect to positive samples, and across all genres. Model 4's macro recall is equal to 0.9392, showing it performed fairly well when considering coverage across the various genre classes without any significant biases. Lastly, the best macro F1 score of 0.9391 demonstrated a balanced approach between precision and recall but not without trade-offs. Where our data is evenly distributed, it is important to consider macro-averaged metrics that ultimately would evaluate across each genre class together.

5.2. Comparative Analysis of Models

Model 1 had only three dense layers and no dropout, and it was underfitting. It reached a test accuracy saturation of 87.6% and was the model with the least F1-score (0.8742), indicating it lacked capacity to generalize elastic temporal and spectral patterns.

Model 2 was an improved architecture that included an additional hidden layer with 20% dropout and validated (92.77%) and test (91.3%) accuracy levels. Dropout was

included in the architecture to reduce overfitting and the optimizer was the Adam optimizer as it provided consistency in the smooth updating of the gradients.

Model 3 was the same architecture as in Model 2 but used SGD as the optimizer. While this model did get an F1-scores that were slightly lower than in Model 2 (0.9091), the training curve showed more fluctuation compared to Model 2 as SGD converged slower and was more sensitive to learning changes.

Model 4 with five hidden layers, 30% dropout, and using RMSprop, has yielded the best results at the total model level. RMSprop with its adaptive approach to learning rates may have improved the model's ability to manipulate lower weight and possible non-stationary gradients, aiding the model to reach a stable appropriate convergence.

5.3. Confusion Matrix Interpretation

The confusion matrix for Model 4 is shown in Figure 8. The matrix exhibits a strong diagonal pre-dominance, in line with the model's ability to correctly define the majority of genres.

- **Classical, Metal, and Pop:** These genres were nearly all recognized correctly, as they exhibit strong features of spectromorphology and unique rhythmic characteristics.
- **Rock vs. Country:** This genre pair exhibited errors likely due to instrumentation overlap and similar tempo ranges.
- **Hip-hop vs. Pop:** While this pair was the lowest confusion of all pairs in the genres, there remains some similarities in regards to characteristics of rhythm and production; particularly with regards to more vigorous characteristics of percussion, and frequency of beat.

These results reflect a real-world challenge of trying to separate genres that possess stylistic and/or fusing characteristics.

5.4. Insights from Training Curves

The training curves (Figures 13 to 16) provide good insight into each model's optimization behavior:

- **Model 1:** Showed early convergence with a small gap between training and validation curves—suggesting underfitting.
- **Models 2 and 3:** Showed modest overfitting, although dropout reduced variance between the training and validation.
- **Model 4:** demonstrated smooth and gradual convergence with continuous improvement and little overfitting, showing good generalization.

The combination of dropout regularization and adaptive learning rate strategies (RMSprop) were critical in the stability of the training and generalization capabilities.

5.5. Genre Clustering and Tempo Insights

A PCA projection (Figure 7) offers a qualitative viewpoint on genre separability in reduced dimensions:

- **Classical and Hip-hop:** shows clear separability probably due to disparate harmonic structures and rhythmic density.
- **Disco and Reggae:** there was partial overlap as disco and reggae genres emphasized strong rhythmic motifs as both genres have a compatible beat frequency.

The tempo analysis (Figure 6) analysis supports the above:

- **High Tempo Genres (Metal, Disco):** These genres are clearly distinguishable due to their fast-paced, rhythm-driven profiles.
- **Low Variability Genres (Pop, Jazz):** These genres had more defined tempo ranges.
- **Wide Tempo Spread (Blues, Classical):** These genres had greater variability, which might have presented some challenges for intra-class predictions.

5.6. Discussion

The results obtained from traditional audio features in conjunction with fully connected networks support our hypothesis: lightweight models can achieve competitive genre classification performance when combined with rich, interpretable features. Our models also yield faster inference times than systems based on CNN or RNN technology, making them suitable for real-time MIR applications.

Furthermore, our method inherently lends itself to a higher level of interpretability based on structured features, PCA visualizations, and genre-based features. These features can be used not only for improving classification methods, but also for providing explanations to end users in consumer music applications.

The examined confusions seen from the confusion matrix suggest work should also be done examining more advanced attention mechanisms or ensembling techniques, to enhance genre boundary boundaries. Additionally, a real-world scenario would benefit from robustness testing with noise, compression, or polyphonic audio samples.

6. Conclusion

In this study, we proposed a framework for automatic music genre classification problem based on deep feed-forward networks with a thoroughly defined audio feature

pipeline. We used the GTZAN data set as a benchmark and demonstrated that interpretable hand-crafted audio features together with carefully devised fully connected bare architectures performed similarly as more complex convolutional or recurrent neural network based models, and without a considerable computation overhead.

Our method is an entire pre-processing and modeling pipeline of a signal: it contains silence trimming, generation of 3 second segments, and 58 domain relevant audio features that were also hand-crafted and professionally extracted, to include MFCCs, chroma vectors, tempo, spectral roll-off, and zero-crossing rate. These audio features represent the temporal, harmonic and rhythmic foundations to musical genre. After a z-score normalization and optional PCA visualization, the data sets we can train the feedforward DNNs on, are well fitted and provide room for generalization.

Of all four architectures tested, Model 4, which has five dense layers, 30% dropout and uses RMSprop for optimization, resulted in the best performance, achieving a test accuracy of 93.92% while also returning macro-average precision, recall, and F1-score of greater than 0.93. These results demonstrate the strength of our architecture’s performance and generalization capacity facilitated by hit or dry-domain regularization techniques and adaptive learning.

In addition to quantitative performance metrics, we are also committed to interpretability through rich visualizations. Confusion matrices, PCA plots, training dynamics, and tempo boxplots all provided us with ways to analyze model behavior at the level of genre specific to the study. We consistently saw genres with strong timbral or tempo characteristics (such as classical, metal and disco) be accurately classified, while small confusion was seen with genres that share stylistic features (such as country, rock and hip-hop). The analysis provided aspects for further model refinement or opportunities for feature engineering.

This study is distinguished by not continuing with the trend of using heavy, black-box CNN architectures for analysis of music. We have demonstrated that with an intentional selection of features and tuning of architecture, fully connected networks can achieve state-of-the-art performance while providing greater transparency, faster inference, and compatibility with lightweight or embedded hardware; thus making our approach easily transferrable to commercial music streaming platforms, mobile recommendation engines, and real-time genre-tagging pipelines.

In short, our findings demonstrate that traditional audio descriptors can effectively be incorporated into a deep learning workflow, allowing a balance between interpretability and performance. Our contribution is to provide a modular, reproducible, and scalable framework for future research in the wider field of Music Information Retrieval, especially in contexts where efficiency and explainability

are equally as important as accuracy in task outcomes.

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